



THE BOXES

Research dossier

A drama set inside NASA Mission Control across the four days of Apollo 13, from the oxygen-tank rupture to splashdown in April 1970.

The research reconstructs the Manned Spacecraft Center in Houston during the four days of the Apollo 13 crisis in April 1970. Engineering teams inside Building 30 managed hardware failures through cathode-ray consoles, paper documentation, and real-time computation facilities. This film observes the operation through the controllers whose pre-launch oversights set the tank explosion in motion. They work inside the closed environment of the control floor as the consequences of their earlier procedural sign-offs unfold in real time.

Depth	Production
Research confidence	77%
Coverage	85%
Status	—
Boxes	13
Evidence fragments	241
Distinct sources	186
Distinct domains	72
Contradictions	0 conflict, 0 context
Research passes	4
Generated	2026-09-09 15:43 UTC
Project id	demo

THE BOXES

What the research found, in plain language, box by box. Every claim below is traceable to the sources listed under it.

PRESS CORPS INFRASTRUCTURE

99%  8 items · 5 domains

Press operations operated out of the Building 1 auditorium adjacent to the control wing. NASA External Relations installed teletype networks, bank payphones, and broadcast camera positions for print and television journalists. Briefings delivered mission reports and operational updates directly to reporters filing real-time copy. Security boundaries separated the working press pool in Building 1 from the operations floor inside Building 30.

- Building on a Mission: The Auditorium and External Relations Office Building - NASA · 2023-09-28
- News Media Handbook · 2005-02-01
- The display/control complex of the Manned Space Mission Control ...
- The Press Site at KSC · 2006-11-29
- Mission Control Center (MCC) - Apollo 13 - Fourth (4th ...
- APOLLO 13 MISSION REPORT SEPTEMBER 1970

OFFICE OF CIVILIAN DEFENSE RELATIONS

98%  12 items · 6 domains

Federal security details and local Houston police departments monitored the aerospace complex perimeter. Heightened domestic tensions and ongoing anti-war protests required active liaison with regional law enforcement agencies. Site access passes were checked at vehicle barricades and entrance lobbies across the base. Security personnel documented external inquiries and safeguarded sensitive technical data throughout the mission.

- Records - Houston Police Department
- [PDF] Police Practices in Houston, Texas - U.S. Commission on Civil Rights
- Records transmittal and receipt from NASA Manned Spacecraft Center to Federal Records Center, GSA, Fort Worth, Texas for Record Group 255
- Historical Documents - Office of the Historian · 2026-06-29
- Building on a Mission: The Houston Mission Control Center - NASA · 2023-09-23
- Research in Vietnam War Records
- Houston Police Department - Public Information Requests
- Vietnam War: The War at Home | National Archives · 2022-06-17
- News Release Archives
- Houston Police Department
- Transparency

REAL-TIME TELEMETRY DATA FLOW · emergent

97%  14 items · 7 domains

Spacecraft telemetry flowed through worldwide Deep Space Network and Manned Space Flight Network stations directly to central mainframes. Antenna installations such as Honeysuckle Creek routed tracking data through Goddard Space Flight Center to Houston. Building 30 computers converted radio-metric and health parameters into display vectors for floor consoles. Wall plotters and individual screens updated in real time as digital streams arrived from overseas dishes.

- MANNED SPACE FLIGHT NETWORK FOR APOLLO - MIT · 2002-10-31
- FLIGHT-CONTROL DATA NEEDS, TERMINAL DISPLAY ... · 2005-06-09
- Apollo Era Technical Description
- Deep Space Network - NASA Science · 2025-01-16
- Integrated telemetry-computer system
- Telemetry and Command (T&C) Systems · 2026-03-13
- https://tda.jpl.nasa.gov/progress_report/42-25/25C.PDF
- NATIONAL AERONAUTICS AND SPACE ADMINISTRATION 1971 15,
- 1970 December 27 ~(C
- [PDF] Real-Time Flight Test Data Distribution and Display
- Network Space Deep The 7, Progress'Report.42-40
- SECTION 7 CRT DISPLAY PAGES
- -

ENGINEERING DIVISION DEMOGRAPHICS

97%  15 items · 5 domains

NASA assembled its Apollo flight operations teams from young civilian engineers, contractor specialists, and military veterans. Many flight controllers were recruited straight out of technical universities and assigned directly to active mission stations. Their service files document specialized system assignments along with recent military flight backgrounds. These young personnel carried full operational authority over individual spacecraft subsystems.

- Information Summaries Astronaut Fact Book - NASA · 2013-04-08
- Astrona · 2008-05-23
- Apollo 13 - NASA · 2023-03-29
- Apollo 13 Archives - NASA
- List of NASA's flight control positions - Wikipedia · 2026-06-21
- Flight controller - Wikipedia · 2026-06-29
- of
- Documentary Spotlights Apollo-Era NASA Flight Controllers – Houston Public Media · 2017-04-15
- Msc Personnel
- Apollo 13 II - NASA · 2023-07-06
- biographical summaries

CONSOLE HARDWARE ARCHITECTURE

96%  16 items · 9 domains

Building 30 housed Mission Operations Control Rooms across its second and third floors. Controllers sat at fixed consoles facing monochrome CRT displays, illuminated button arrays, and pneumatic carrier delivery tubes. Equipment bays in the complex fed telemetry and operational configurations into individual station monitors. These physical layouts anchored each flight operations position to specialized console wiring.

- Apollo Mission Control Center Restoration - NASA · 2026-04-08
- NASA's restored Apollo Mission Control is a slice of '60s life, frozen in amber - Ars Technica · 2025-05-16
- Apollo Mission Control Photos - Space Center Houston · 2022-09-23
- MOCR - Manned Spaceflight Operations Association · 2018-03-13
- Orbiter Displays Reference Guide (OI-34)
- AS-508 MISOO FIU A IN
- MCC Operational Configuration Mission J1 AS-510/SC112/LM1C Apollo 15
- COMMAND AND SERVICE MODULE CONTROLS AND ...
- mission operations center: Topics by Science.gov
- CRT Reference Summary OI-27
- Microsoft Word - Final Drft, MCC 2010-12-30
- [PDF] Apollo Mission Control Restoration - NASA
- GitHub - johnmknigh/MOCR-Console-Replica-Suite: Period ...
- The display/control complex of the Manned Space Mission Control ...

COMMUNICATIONS LOOP ACOUSTICS

96%  17 items · 11 domains

Mission communications were divided across dedicated flight director, Capcom, Trench, and backroom voice loops. Ground controllers maintained real-time flight chronologies while speaking over air-to-ground channels to the spacecraft crew. Audio fidelity varied between high-bandwidth local circuits and long-distance tracking relays. The recorded exchanges capture distinct pacing, concise acknowledgments, and overlapping technical speech.

- Apollo 13, transcription du vol
- Mission Control Audio Channel
- Apollo 13 in Real Time
- Ben Feist Interview on Apollo 13 Audio | Everyday Astronaut · 2020-11-23
- Apollo 13 Transcripts
- How Mission Control saved Apollo 13 | BBC Sky at Night Magazine · 2026-06-15
- Mission Transcripts: Mercury, Gemini, and Apollo - NASA · 2026-06-15
- NASA Audio Highlight Reels, Soundbites and Launch Sounds
- Apollo 13 Audio

- NASA, university researchers release 19000 hours of ... · 2018-07-25
- Apollo 11 : NASA : Free Download, Borrow, and Streaming · 2010-12-07
- Recovering Audio from the Apollo Moon Missions
- Apollo 11 - Technical Air-to-Ground Voice Transcription

CLEAR LAKE COMMUTING

95%  12 items · 7 domains

The Manned Spacecraft Center was constructed on open flatland twenty-five miles southeast of downtown Houston near Clear Lake. Flight controllers commuted along state highway access routes and local access roads to reach the facility gates. Surface parking lots around the operations buildings held late-1960s personal sedans and domestic vehicles. Security gates controlled vehicular access from the public roadway network.

- 309 Clear lake Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 19 Clear lake area, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Clear Lake - Police - City of Houston
- 5 Lct area, 1970 s Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- ROAD SURFACES AROUND NASA, NASA Technology Images — PICRYL - Public Domain Media Search Engine
- 60 Years Ago: NASA Selects Houston as Site for New Manned Spacecraft Center - NASA · 2023-08-30
- Microsoft Word - If99-6723_final-AVT-152-005_NASA_Experience_w-Pogo_in_Human_Spaceflight_Vehicles_Final.doc
- 104 Clear lake, Texas Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 100 Clear lake, Houston Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- City of Houston eGovernment Center > About Houston > NASA
- CLEAR LAKE No. 81
- MAP OF THE CLEAR LAKE AREA

TECHNICAL CRISIS VOCABULARY

94%  10 items · 6 domains

Flight controllers relied on an extensive vocabulary of telemetry acronyms, orbital mechanics shorthand, and operational callouts. Technical glossaries defined standardized abbreviations for systems like the Lunar Module malfunction procedures and guidance interfaces. Controllers used terms such as GUIDO, Trench positions, and specific abort modes in daily console communication. These compact codes compressed complex life-support and propulsion states into brief spoken sentences.

- [PDF] - 3 GLOSSARY OF ACRONYMS AND ABBREVIATIONS - Ibiblio
- GLOSSARY OF TERMS · 2010-01-27
- Appendix B: Acronyms, Abbreviations, and Symbols - NASA · 2026-04-22
- Apollo 13 LM Malfunction Procedures
- [PDF] _.-_m MANNED SPACECRAFT CENTER - NASA
- NASA-Apollo-Mission-Control-Acronyms. ...
- Space Transportation System and Associated Payloads: Glossary, Acronyms, and Abbreviations
- Space Flight Technical Language Handbook
- APOLLO 13 MISSION REPORT SEPTEMBER 1970

FACILITY LIGHTING PROTOCOLS

94%  10 items · 9 domains

The Mission Operations Control Room was kept in deep ambient dimness to preserve readability on the overhead projection screens. Narrow spotlights illuminated console desks while keeping foot-candle values on the main floor extremely low. Fluorescent ceiling fixtures remained turned off during active tracking passes. The resulting darkness focused all visual attention on glowing projection panels and console cathode-ray monitors.

- PPFD to Foot-candle Reference Table
- Foot-candle - Wikipedia · 2026-03-08
- FOOTCANDLE LIGHT GUIDE - Lighting Design Lab · 2013-07-29
- Foot Candle Calculator - Lighting Requirements & Illuminance ...
- https://www.usmd.edu/usm/adminfinance/Facilities/forms/interior_lighting_fluorescent.pdf
- Lighting Illumination Calculator - everycalculators.com

- Apollo Flight Controller 101: Every console explained - Ars Technica · 2019-12-25
- 3.PDF
- 20F Technology Committee On-Screen Light Measurement ...

MANUAL CALCULATION TOOLS · emergent

89%  14 items · 5 domains

Controllers relied on manual calculation tools when primary telemetry and computer systems suffered dropouts. Engineers kept five-inch Pickett slide rules, printed graph paper, and grease pencils directly on console desks. Mechanical desk calculators and reference manuals supported manual power budget tracking and trajectory verifications. These analog devices confirmed navigation solutions alongside electronic mainframe calculations.

- Apollo Command Module Block II Primary Guidance, Navigation, and Control System Manual · 2018-07-27
- Quasi Official Apollo 13 Paper September 8.ppt · 2009-09-08
- - · 2004-09-22
- Slide Rule, 5-inch, Pickett N600-ES, Apollo 13 | National Air and Space Museum
- Don't Know What a Slide Rule is For | National Air and Space Museum
- STUDY OF CONTROL COMPUTERS FOR ...
- AS-508 MISOO FIU A IN
- E-2448 Rev 1, Users' Guide to Apollo GN&CS Major Modes and Routines
- 1970 Slide Rule
- Slide rule vintage hi-res stock photography and images
- ORBITAL TESTING REQUIREMENTS FOR GUIDANCE AND CONTROL DEVICES

CONTROL ROOM CONSUMABLES

62%  38 items · 13 domains

Console desks were crowded with standard commercial tobacco packs, filled glass ashtrays, and disposable paper coffee cups. Food services and on-site vending machines supplied coffee, pastries, and packaged snacks to staff working uninterrupted shifts. Takeout boxes and paper cartons accumulated directly on active operational surfaces throughout the crisis. Standard institutional provisions sustained personnel who remained locked inside the facility.

- 93 Space food, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 120 Food containers, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Food Services - NASA · 2026-09-04
- NASA Facts
- [PDF] LYNDON B. JOHNSON SPACE CENTER
- Apollo Mission Control Photos - Space Center Houston
- 130 Food container, Crew Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 132 Food container, Activities Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 223 Spacecrew supplies Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Apollo Mission Control Restoration - NASA
- APOLLO EXPERIENCE REPORT
- Apollo 13 Flight Journal - Mission Documents
- N76 12690 · 2004-01-23
- 163 Food container, Nasa Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 100 Space food, Food Images: PICRYL - Public Domain Media Search Engine Public Domain Search

SHIFT ROTATION ATTIRE

58%  27 items · 12 domains

Archival photographs record the progressive physical exhaustion of the flight control staff across eighty-four hours of duty. Controllers worked continuously in short-sleeve button-down shirts, neckties, and slacks. Images capture loosened ties, wrinkled fabric, and heavy perspiration as shifts lengthened. Ash fell onto clothing and work surfaces as personnel smoked cigarettes directly at their console stations.

- Mission Control, Houston, April 13, 1970 - NASA · 2015-04-13
- Apollo 13: The Successful Failure · 2020-04-06
- Apollo 13 - Wikipedia · 2026-08-13

- Apollo 13
- File:1970 Mission Control Apollo 13.jpg - Wikimedia Commons
- How Mission Control saved Apollo 13 - BBC Sky at Night Magazine
- PHOTOGRAPHS WRITTEN HISTORICAL AND ...
- 165 Flight controllers, Astronauts Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 992 Controllers Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- 252 Consoles, Control Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- Apollo 13, NASA Apollo program. NASA public domain image... — PICRYL - Public Domain Media Search Engine
- Project Apollo Archive's albums | Flickr
- skandalon
- 43 Flight clothing, Astronauts Images: PICRYL - Public Domain Media Search Engine Public Domain Search
- I found an archive of every picture taken by the astronauts ...

SUPPORT STAFF TRENCH DYNAMICS · emergent

26%  32 items · 13 domains

The front-row Trench worked in direct communication with Staff Support Rooms located immediately behind the main hall. These backrooms held teams of specialized engineers, trajectory specialists, and computing technicians. The Real-Time Auxiliary Computing Facility processed trajectory figures and fed updated parameters to the main console operators. Hierarchical task division ensured backrooms generated raw figures while front-row controllers made final calls.

- Apollo experience report - real-time auxiliary computing facility development · 2002-04-17
- Mission Control, Houston · 2006-05-01
- Going boldly: Behind the scenes at NASA's hallowed Mission Control Center - Ars Technica · 2012-11-14
- The Trench: The People at Mission Control | American Experience | Official Site | PBS · 2017-12-28
- Diagram of NASA Mission Control | National Air and Space Museum
- Apollo Mission Control Center Restoration · 2026-04-08
- The newly restored Apollo Mission Control Room : r/nasa
- Background Data (Process Overview)
- FLIGHT-CONTROL DATA NEEDS, TERMINAL DISPLAY ...
- MANNED SPACE FLIGHT NETWORK FOR APOLLO - MIT
- MCC Operational Configuration Mission J1 AS-510/SC112/LM1C Apollo 15 · 2002-02-05
- Mission Operations Directorate New Employee Orientation
- Apollo 13
- Apollo 13 (film)
- Apollo 13, We Have a Solution: Part 2 - IEEE Spectrum · 2021-07-25

CONTRADICTIONS

Where two sources disagree. Similarity alone never decides this; each pair below was read and judged.

None found.

REFERENCE REEL

The strongest material, cut into a sequence a director can read.

00:00 Establishing Mission Control

Archival materials document the control room environment during Apollo 13 in Houston.

- Mission Control, Houston, April 13, 1970 - NASA · nasa.gov
- Apollo Mission Control Center Restoration - NASA · nasa.gov · 2026-04-08
- Going boldly: Behind the scenes at NASA's hallowed Mission Control Center - Ars Technica · arstechnica.com · 2012-11-14
- How Mission Control saved Apollo 13 | BBC Sky at Night Magazine · skyatnightmagazine.com · 2026-06-15

00:30 Multi-Channel Communications

Restored multi-track audio channels record simultaneous conversations across flight controller consoles.

- Mission Control Audio Channel · apolloinrealtime.org
- Ben Feist Interview on Apollo 13 Audio | Everyday Astronaut · everydayastronaut.com · 2020-11-23
- Apollo 13 in Real Time · apolloinrealtime.org

01:00 Air-to-Ground Crisis Logs

Technical flight transcripts record critical communications following the oxygen tank failure.

- Apollo 13, transcription du vol · capcomespace.net
- Apollo 13 Transcripts · apollojournals.org
- Apollo 13 Audio · dn801309.us.archive.org

01:30 Extended Crisis Strain

Real-time records capture the extended four-day rescue operation through splashdown.

- Apollo 13 in Real Time · apolloinrealtime.org
- Apollo 13 - NASA · nasa.gov

DEPARTMENTS

The same boxes, grouped by which crew briefs off them, with what each one actually contains.

Script

COMMUNICATIONS LOOP ACOUSTICS · 17 items

Mission communications were divided across dedicated flight director, Capcom, Trench, and backroom voice loops. Ground controllers maintained real-time flight chronologies while speaking over air-to-ground channels to the spacecraft crew. Audio fidelity varied between high-bandwidth local circuits and long-distance tracking relays. The recorded exchanges capture distinct pacing, concise acknowledgments, and overlapping technical speech.

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MANUAL CALCULATION TOOLS · 14 items

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REAL-TIME TELEMETRY DATA FLOW · 14 items

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Casting

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Art direction

CONSOLE HARDWARE ARCHITECTURE · 16 items

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PRESS CORPS INFRASTRUCTURE · 8 items

Press operations operated out of the Building 1 auditorium adjacent to the control wing. NASA External Relations installed teletype networks, bank payphones, and broadcast camera positions for print and television journalists. Briefings delivered mission reports and operational updates directly to reporters filing real-time copy. Security boundaries separated the working press pool in Building 1 from the operations floor inside Building 30.

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Cinematography

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Locations

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PRIOR ART

65 existing films surveyed for a similar premise, seeded from: space crisis, mission control, engineers, survival, 1970s.

Apollo 13 (1995)

exploited mechanical flaw in the oxygen system · victim · triumphant · professional duty overcomes disaster · crew rescued after safe splashdown

similarity 0.72 · <https://www.themoviedb.org/movie/568>

Salyut-7 (2017)

exploited electrical flaw in station hardware · investigator · heroic · loyalty to comrades redeems political coercion · station restored to operational capability

similarity 0.66 · <https://www.themoviedb.org/movie/438740>

The Right Stuff (1983)

exploited psychological flaw in institutional hubris · bystander · satirical · individual audacity transcends bureaucratic manipulation · pioneering era closes with success

similarity 0.66 · <https://www.themoviedb.org/movie/9549>

The Real Right Stuff (2020)

exploited historical record shipped as archival media · bystander · documentary · collective effort secures a national achievement · historical missions conclude successfully

similarity 0.64 · <https://www.themoviedb.org/movie/753325>

First Man (2018)

exploited emotional flaw of unexpressed grief · victim · somber · private sorrow finds release through professional endurance · mission completed with silent personal reconciliation

similarity 0.64 · <https://www.themoviedb.org/movie/369972>

The Martian (2015)

exploited scientific ingenuity shipped as standard equipment · victim · optimistic · relentless problem-solving affirms human resilience · stranded astronaut successfully retrieved

similarity 0.63 · <https://www.themoviedb.org/movie/286217>

Hidden Figures (2016)

exploited systemic bias flaw in workplace hierarchy · investigator · inspiring · competence forces institutional integration · ground calculations validate the orbital trajectory

similarity 0.62 · <https://www.themoviedb.org/movie/381284>

Love (2011)

exploited psychological isolation flaw in prolonged confinement · victim · melancholic · alienation deepens into surreal acceptance · isolated subject achieves transcendence

similarity 0.58 · <https://www.themoviedb.org/movie/54320>

First Man into Space (1959)

exploited biological mutation flaw caused by cosmic radiation · perpetrator · tragic · reckless ambition brings monstrous ruin · dying monster provides crucial scientific data

similarity 0.56 · <https://www.themoviedb.org/movie/28955>

The Devil's Mouth (2026)

exploited physical vulnerability flaw in a trapped environment · victim · dread · survival pressure fractures ethical solidarity · diminished survivors escape the predator

similarity 0.55 · <https://www.themoviedb.org/movie/1481343>

Santo vs. the Martian Invasion (1967)

exploited alien invasion feature shipped as advanced weaponry · investigator · camp · physical heroism preserves terrestrial security · alien invaders suffer total defeat

similarity 0.55 · <https://www.themoviedb.org/movie/80766>

It All Adds Up - The Making of Hidden Figures (2017)

exploited promotional commentary shipped as retrospective access · bystander · celebratory · historical recovery honors overlooked labor · filmmakers validate historical relevance

similarity 0.55 · <https://www.themoviedb.org/movie/1533402>

Where this stands, and where it does not yet

Originality is claimed only relative to the titles above, never absolutely.

The narrative adopts the perpetrator point of view inside Mission Control, focusing on flight controllers whose previous procedural oversights caused the technical failure.

None of the listed titles present the disaster strictly through the culpability of the ground controllers.

checked against: Apollo 13, Salyut-7, The Right Stuff, The Real Right Stuff, First Man, The Martian, Hidden Figures, Love, First Man into Space, The Devil's Mouth, Santo vs. the Martian Invasion, It All Adds Up - The Making of Hidden Figures

→ Frame the central flight controller as the engineer who signed off on the compromised oxygen tank specifications.

The narrative operates as an institutional tragedy ending in the irreversible destruction of administrative careers despite the survival of the flight crew.

None of the listed films conclude with bureaucratic self-destruction alongside a successful rescue.

checked against: Apollo 13, Salyut-7, The Right Stuff, The Real Right Stuff, First Man, The Martian, Hidden Figures, Love, First Man into Space, The Devil's Mouth, Santo vs. the Martian Invasion, It All Adds Up - The Making of Hidden Figures

→ Show ground command securing splashdown while an internal audit terminates their program authority.

The narrative adopts a tone of relentless bureaucratic dread, focusing exclusively on ground support personnel paralyzed by statistical inevitability.

None of the listed historical or space flight films use an sustained tone of pure procedural dread without heroic relief.

checked against: Apollo 13, Salyut-7, The Right Stuff, The Real Right Stuff, First Man, The Martian, Hidden Figures, Love, First Man into Space, The Devil's Mouth, Santo vs. the Martian Invasion, It All Adds Up - The Making of Hidden Figures

→ Confine the perspective entirely to telemetry analysts convinced their models predict crew asphyxiation.

RESEARCH LOG

What each pass searched, found, and did next. Process detail, not required reading.

PASS 000

114 sources · 133 fragments · 40 via Extract

coverage 0% → 82% confidence 0% → 77%

CONSOLE HARDWARE ARCHITECTURE — Apollo Mission Operations Control Room MOCR console schematics Philco Ford CRT pneumatic tube; NASA MSC Building 30 MOCR console hardware display controls layout technical report

COMMUNICATIONS LOOP ACOUSTICS — Apollo 13 flight director loop capcom audio recordings transcript April 1970; NASA Apollo 13 internal communications voice loop trench backroom audio tapes archive

ENGINEERING DIVISION DEMOGRAPHICS — Apollo 13 flight controllers roster biographies educational background military service; NASA MSC Flight Operations Division personnel demographics flight controllers 1970

SHIFT ROTATION ATTIRE — NASA Mission Control Apollo 13 high resolution archival photographs controllers MOCR April 1970; Apollo 13 mission control room shift rotation candid photos controllers clothing rumpled

CONTROL ROOM CONSUMABLES — NASA Mission Control center Building 30 cafeteria vending coffee brands tobacco 1960s 1970; Apollo 13 mission operations control room ashtrays coffee cups takeout food archival photos

TECHNICAL CRISIS VOCABULARY — Apollo 13 telemetry acronyms abbreviations flight controller jargon mission operational terms; NASA Apollo flight controller terminology life support EPS EECOM telemetry shorthand handbook

FACILITY LIGHTING PROTOCOLS — MSC Building 30 Mission Operations Control Room lighting levels foot-candles display illumination specifications; Apollo MOCR rear projection screens ambient illumination lighting protocols technical manual

CLEAR LAKE COMMUTING — NASA Manned Spacecraft Center Clear Lake employee parking lot gate security vehicles 1970; Clear Lake Houston NASA Road 1 commute personnel parking cars April 1970 historical photos

PRESS CORPS INFRASTRUCTURE — MSC Building 1 auditorium news center Apollo 13 press corps floor plan teletype; Apollo 13 press room television camera positions NASA Manned Spacecraft Center newsroom setup 1970

OFFICE OF CIVILIAN DEFENSE RELATIONS — NASA Manned Spacecraft Center security procedures Vietnam War antiwar protests 1970; MSC Houston perimeter security police civil defense coordination NASA complex April 1970 records

→ investigate CONTROL ROOM CONSUMABLES (score 0.31)

PASS 001

36 sources · 34 fragments · 12 via Extract

coverage 82% → 82% confidence 77% → 76%

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SHIFT ROTATION ATTIRE — Apollo 13 Mission Operations Control Room high resolution archival photos controllers April 1970; NASA flight controllers clothing shirts ties fatigue Apollo 13 photographs shift changes

CONTROL ROOM CONSUMABLES — Apollo 13 Mission Control coffee cups cigarette brands food containers April 1970; NASA MOCR consoles food drink ashtrays vending snacks Apollo missions archive photos

SUPPORT STAFF TRENCH DYNAMICS — NASA Apollo Mission Control Staff Support Room backroom layout procedures trench; Apollo 13 RETRO FIDO backroom trajectory calculation procedures support staff organizational chart

→ investigate SUPPORT STAFF TRENCH DYNAMICS (score 0.26)

PASS 002

38 sources · 32 fragments · 12 via Extract

coverage 82% → 83% confidence 76% → 75%

opened: MANUAL CALCULATION TOOLS, SUPPORT STAFF TRENCH DYNAMICS

CONTROL ROOM CONSUMABLES — Apollo 13 Mission Operations Control Room MOCR coffee cups cigarettes food trash photographs April 1970; NASA MSC Building 30 Mission Control consumables vending machines ashtrays brands 1970

SUPPORT STAFF TRENCH DYNAMICS — NASA Apollo Staff Support Room backroom layout procedures Trench FIDO GUIDO flight dynamics; Apollo 13 Mission Operations Control Room Staff Support Room organizational hierarchy communication loop

MANUAL CALCULATION TOOLS — Apollo 13 mission control slide rules calculators grease pencils trajectory manual calculation tools; NASA MOCR Apollo flight controller slide rule model Friden calculator graphing paper inventory 1970

→ investigate SUPPORT STAFF TRENCH DYNAMICS (score 0.26)

PASS 003

41 sources · 33 fragments · 12 via Extract

coverage 83% → 85% confidence 75% → 76%

opened: REAL-TIME TELEMETRY DATA FLOW, SUPPORT STAFF TRENCH DYNAMICS

CONTROL ROOM CONSUMABLES — Apollo 13 Mission Operations Control Room food coffee cigarettes photos; NASA Mission Control Center building 30 consumables vending coffee cups 1970

SUPPORT STAFF TRENCH DYNAMICS — Apollo 13 Staff Support Room backroom flight dynamics trench organizational hierarchy; Apollo mission control SPAN Trench backroom procedures calculations documentation

REAL-TIME TELEMETRY DATA FLOW — Apollo telemetry data flow MSFN Goddard Real-Time Computer Complex MOCR displays; NASA Manned Spacecraft Center Real Time Computer Complex RTCC CRT display generation 1970

→ investigate SUPPORT STAFF TRENCH DYNAMICS (score 0.26)